

Power Bi Project Cleaning Summary Note Disclaimer

Prepared by: Diaa Aldein Alsayed Ibrahim Osman

Duplicate Removal

- In `Dim\_Orders`, 4 out of 5 duplicate rows for Order ID `AZ-2014-8495612` were removed.
- In `Fact\_Order Details`, 3 out of 4 duplicate rows for Order ID `AZ-2012-8977452` were removed.

Error Handling and Value Correction

1. Table Dim_Orders:

- Corrected errors in the *Customer Key* ("13" → 13) and *Ship Date* ("text" → "7/31/2012" after investigation).

2. Table Fact_Order Details:

- Fixed errors in:
 - **Product Key** ("546" → 546) and **Sub-Category Key** ("14" → 14)
- **Sales**: Replaced ("Text" with 194.25), calculated using the unit price from a related order.
 - Used the **Product Key 845** with **Order ID AZ-2012-3906554** to determine the unit price:
Unit Price = $518 / 8 = 64.75$.
 - Applied this unit price to the **Product Key 845** with **Order ID AZ-2012-1952755**:
Sales = $64.75 \times 3 = 194.25$
 - Both orders share the same year and shipping mode.
- **Profit**: Replaced ("Text" with 23.625), derived from recalculated cost and sales data.
 - Used the **Product Key 845** with **Order ID AZ-2012-3906554** to determine the total cost:
Total Cost = Sales (518) – Profit (63) = 455.
Cost per Unit = Total Cost (455) / (8) = 56.875.
 - Applied this cost per unit to the **Product Key 845** with **Order ID AZ-2012-1952755**:
Total Cost = $56.875 \times 3 = 170.625$
 - Profit = Sales (194.25) – Total Cost (170.625) = 23.625.
 - Both orders share the same year and shipping mode.

Resolving Primary Key Issues

- In **Table Dim_Product**, duplicate primary keys were addressed by grouping records and combining product names with a " / " delimiter.

Order ID Duplication in `Fact\_Order Details`

- Kept duplicate Order IDs, as each represents a distinct product within the same order.

Null Value Handling

- Identified six null **Ship Date** values in **`Fact\_Order Details`** (five in 2011, one in 2013).
- Since these rows lacked corresponding data in **`Dim\_Orders`**, they were filtered out to prevent issues in modeling and time series analysis.